LGCRS: LLM-Guided Representation-Enhancing for Conversational Recommender System

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Abstract. Conversational recommender systems primarily focus on acquiring user preferences through real-time interactions. However, their effectiveness in modeling user preferences is constrained by the amount of conversational information. Existing conversational recommendation methods typically leverage external knowledge graph to enhance recommendation performance, potentially overlooking valuable textual information related to users and items. With significant natural language understanding capabilities, large language models (LLMs) have the potential to improve the performance of conversational recommendation task by improving the quality of text representations. However, the incorporation of raw textual information may inadvertently introduce noise to recommendations. Additionally, the semantic gap between the textual information and the collaborative signals could potentially degrade the performance of the conversational recommender system. To tackle the aforementioned challenges, we propose a LLM-guided representation-enhancing method for conversational recommender system, which fuses collaborative signals and semantic information to improve recommendation performance and generate high-quality responses. Specifically, we leverage LLMs to refine item profiles while reducing noise in text information, construct fused item representations across multiple aspects, and align the LLMs-enhanced semantic representation with the CF-side rational representation through regularization terms in the training stage. Experiments on two publicly available conversational recommendation datasets show that our method exhibits better performance in both recommendation and conversation tasks.

Keywords: Conversational Recommender System · Large Language Models · Representation Learning

1 Introduction

In recent years, conversational recommender system (CRS) [6] has become an emerging research topic, which aims to obtain users' real-time interests through dialogue to provide high-quality recommendation results. CRSs focus more on

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extracting user preferences from users' natural language and explicit feedback during dialogue than traditional recommender systems.

Due to users' limited patience, CRSs frequently encounter brief conversations and insufficient contextual information [6]. To address this issue, existing research often incorporates external data sources to enrich contextual information: KBRD [1] and KGSF [23], incorporate knowledge graphs into the recommendation process to refine item representations; RevCore [12] integrates user reviews into conversations to enhance contextual information. However, previous approaches have primarily relied on external knowledge graphs to refine item representations, thereby overlooking the valuable textual information associated with items. The lack of the textual information may restrict the informativeness within the learned representations [14]. This motivated us to develop a method to enhance the understanding of items by integrating item-relevant text information into conversational recommender systems.

In recent years, prominent LLMs like GPT-4 [13] and LLaMA [17] have showcased remarkable capabilities in the domain of natural language processing. The phenomenon has sparked significant interest among researchers, who are actively exploring the capabilities of LLMs in handling textual information within recommender systems [5]. This motivates us to explore the utilization of LLMs for generating meaningful textual representations of items, aiming to enhance CRS performance. However, the incorporation of item-related textual information into conversational recommender systems presents two distinct challenges: (1) Unexpected noise in item-related text information can reduce the effectiveness of the learned item representation. (2) The semantic gap between the LLMs-enhanced semantic representation and the CF-side rational representation of item may decrease the performance of recommendation task.

To address the above challenges, we propose a LLM-Guided Representation-Enhancing method for Conversational Recommender System (LGCRS). Specifically, we first utilize LLMs to refine user reviews and generate textual item profiles. The core idea is to use manually designed prompts to guide LLMs in generating accuracy item profiles while reducing inherent noise in the raw textual information. In this way, we can introduce item semantic information into the CRS, thereby enhancing the system's understanding of the item. Subsequently, we employ distinct encoders to encode the conversational context, external item-oriented knowledge graph and LLMs-enhanced item profile, respectively. By utilizing regularization terms to reduce the semantic gap between the LLMs-enhanced semantic representations and the CF-side rational representations, along with employing a gate layer to fuse these two representations, we obtain more accurate item representations. Based on the fused item representations, we develop a recommendation module for making accurate recommendations and a conversation module for providing high-quality responses. Overall, our contributions can be summarized as follows:

- We introduce a conversational recommendation method that harnesses the textual item profiles refined by large language models to enhance item representations and improve recommendation performance.

- We introduced two regularization terms to reduce the semantic gap between the LLMs-enhanced semantic representation and the CF-side representation.
- Extensive experiments on two widely used conversational recommendation datasets demonstrate that our proposed model achieves concrete improvements in both recommendation accuracy and dialogue quality.

2 Related Work

2.1 Conversational Recommender System

In recent years, with the rapid advancement of dialogue systems, interactive conversations with users have become an attractive method for capturing dynamic user intents and preferences. CRSs typically consist of three main components: the conversation module, the policy module, and the recommendation module [6]. These components may have different emphases depending on the specific CRS type.

Attribute-based conversational recommender systems [9,10,16] primarily focus on constructing the policy module with the goal of making effective recommendations in the fewest dialogue turns possible. These models often employ reinforcement learning techniques to train the dialogue manager, aiming to maximize expected rewards over a longer time horizon. Consequently, they tend to have a simplified conversation module, such as using fixed templates with slots to populate recommendation results and generate the system's response text. These approach aim to streamline the recommendation process and provide users with relevant suggestions efficiently.

End-to-end conversational recommender systems [1,11,12,21,23,24] prioritize the delivery of smooth and coherent text responses to users. These systems seamlessly incorporate information about recommended items into their replies, thereby enhancing the interpretability and acceptability of the recommendations. In such models, the policy module is implicitly integrated within the conversation module. These approaches are designed to offer more natural and engaging conversational experiences while ensuring that the recommended items are presented in a user-friendly and comprehensible manner.

This paper extends the second category of research by incorporating item-related textual information into CRSs. We leverage large language models to reduce noise in textual information and build item semantic representations. By integrating the LLMs-enhanced semantic representations with the CF-side rational representations, we construct meaningful representations and improve the recommendation performance of the model.

3 Preliminaries

Formally, let u denotes a user from user set $\mathcal{U}$, i denotes an item from item set $\mathcal{I}$. A conversation C consists of a list of utterances, denoted by $C = \{d_t\}_{t=1}^{n}$, in which each utterance d_t is a conversation sentence at the t-th turn.

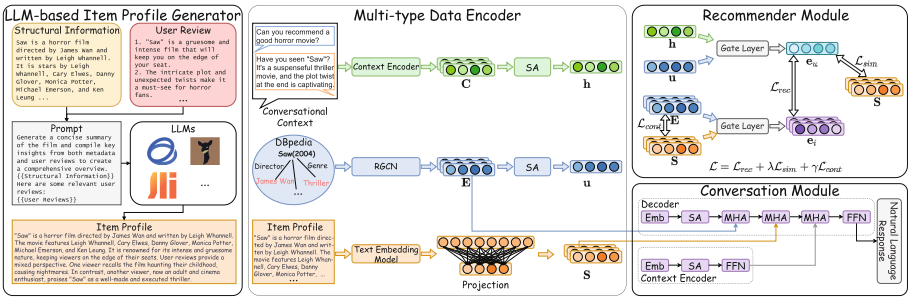

Fig. 1. The overview of our model LGCRS. First, we use LLM-based Item Profile Generator to generate accurate item profile. Then, Multi-type Data Encoder encodes conversational context, external item-oriented knowledge graph and LLMs-enhanced item profile, respectively. Finally, based on the learned representation, we develop a recommendation module for making accurate recommendations and a conversation module for providing high-quality responses. "SA", "MHA" denotes *self-attention* and *multi-head attention*, respectively.

To estimate the user's preferences based on their interactions, the CRS aggregates the utterances d_t in the conversation C and learns a vector $\mathbf{h}$ known as the conversation context. In the t-turn, the recommendation component employs estimated user preferences to generate a candidate item list $[i_1, i_2, \cdots, i_n]$ from the full range of available items $\mathcal{I}$. The conversation component then generates a response discourse that denotes the recommended items to the user.

In addition to the conversation history, previous work has introduced external data to enhance the performance of CRS. External data for CRS typically falls into two categories: structured external data, such as knowledge graphs, and unstructured external data, such as user-written reviews. The knowledge graph $\mathcal{G}$ is formed by a set of entities, denoted by $\mathcal{E}$, and a set of relations, denoted by $\mathcal{R}$. The entity set comprises all the items (*i.e.*, $\mathcal{I}$ is a subset of the entity set $\mathcal{E}$) and other item-related entities. Regarding each user, their interactive entity list $[e_1, e_2, \cdots, e_n]$ is denoted by all entities that have been mentioned in their conversation history C_u. Online reviews can be represented as a document consisting of sentences $R = \{r_1, r_2, \cdots\}$.

Based on the symbol definitions mentioned above, the goal of the CRS is to precisely furnish users with recommended items while generating intelligible responses, utilizing variety of contextual data such as conversational history C, knowledge graph $\mathcal{G}$, user reviews R, and other pertinent information.

4 Methodology

In this section, we introduce our proposed LLM-guided Representation-Enhancing method for Conversational Recommender System (LGCRS). Specifically, we first introduce our method for refining noisy user review information by leveraging a large language model generate item profiles. Secondly, we present

the process of encoding the conversational context, external knowledge graph and LLMs-enhanced item profile. Finally, we discuss how to leverage the learned user representation and item representation for recommendation and dialogue generation (Fig. 1).

4.1 LLM-Based Item Profile Generator

In this section, we will introduce textual information such as user reviews to generate item profiles. However, employing raw item text attributes and user reviews as item profiles may introduce unexpected noise. We believe that by manually designed prompts, we can harness the powerful natural language processing capabilities of LLMs to generate accurate item profiles. Therefore, we propose an item profile generation paradigm based on large language models, which refines item-related textual information while reducing inherent noise.

Specifically, we integrate reasoning process into the LLM by incorporating carefully designed task prompt T to mitigate illusions and improve the quality of generated output. The task prompt clearly specifies its role in generating item profiles from user reviews by accurately specifying the input-output content and format. By combining manually designed task prompts T with raw item profile prompts Q_i, we can leverage LLM to generate accurate profiles. The specific process is outlined as:

$$\begin{aligned} \mathcal{P}_i &= \text{LLMs}(T, Q_i), \\ Q_i &= f_i(t_i, \mathcal{N}^i, R^i), \end{aligned} \tag{1}$$

where f_i represents the function that combines the item text attributes into a single string, t_i is the item title, $\mathcal{N}^i$ represents the one-hop neighborhood node set of the item in the knowledge graph, and R^i is the set of user reviews on the item. By combining structural item descriptions based on the $\mathcal{N}^i$ and user reviews R^i, our prompts Q_i provide precise information for LLMs, ensuring that the generated item profile accurately reflect users' general perspectives.

4.2 Multi-type Data Encoder

Context Encoder. The conversation history $C = \{d_t\}_{t=1}^{n}$ is a series of short sentences d_t communicated between the user and the system during a session, describing their informational interaction to achieve the recommendation goal. In order to obtain a representation of the conversation history, we use a standard Transformer to encode it:

$$\mathbf{C} = \text{Transformer}([d_1; d_2; \cdots; d_t]), t \in [1, n], \tag{2}$$

where $[\cdot;\cdot]$ signifies the concatenation operation, and $\mathbf{C}$ represents the contextual word representation obtained from the top layer of Transformer. Subsequently, we employ a self-attention layer to consolidate the acquired contextual word representations into a representation for the conversation history:

$$\begin{aligned} \mathbf{h} &= \mathbf{C} \cdot \alpha_w, \\ \alpha_w &= \text{softmax}(\boldsymbol{b}_c^T \tanh(\mathbf{W}_c \mathbf{C})), \end{aligned} \tag{3}$$

where $\mathbf{W}_c$ and $\boldsymbol{b}_c$ denote the learnable parameter matrix and vector of the self-attention layer.

Knowledge Graph Encoder. To model the associations between items, we introduce an item-oriented knowledge graph denoted as $\mathcal{G}$, with the aim of enriching item-related information. The graph comprises a set of entities $\mathcal{E}$ and a set of relationships $\mathcal{R}$. In this study, we utilize DBpedia as our chosen item-oriented knowledge graph. Triplets in DBpedia can be represented by $\langle e, r, e' \rangle$, where $e, e' \in \mathcal{E}$ are entities and r is a relation from the relation set $\mathcal{R}$. Considering that the inter-entity relations of item-oriented knowledge graphs may have a considerable effect on understanding user preferences, we adopt RGCN [15] to encode knowledge graphs. The process of learning the representations of the next layer can be formalized as follows:

$$n_e^{(l+1)} = \sigma(\sum_{r \in \mathcal{R}} \sum_{e' \in \mathcal{E}_e^r} \frac{1}{Z_{e,r}} \mathbf{W}_r^{(l)} n_{e'}^{(l)} + \mathbf{W}_s^{(l)} n_e^{(l)}), \tag{4}$$

where $n_e^{(l)}$ is the embedding of e at the l-th layer, $\mathcal{E}_e^r$ denotes the neighboring nodes for e with relation r, $\mathbf{W}_r$ is a learnable relation-specific matrix, $\mathbf{W}_s$ is a learnable matrx for transforming the node self-embedding and $Z_{e,r}$ is a normalization factor.

Item Profile Encoder. Utilizing the generated item profile $\mathcal{P}_i$ in Sect. 4.1, we can compute the semantic representation s_i of the item through the following procedure:

$$s_i = \mathcal{T}(\mathcal{P}_i), \tag{5}$$

where $\mathcal{T}$ represents the text embedding model, a technology proven to effectively transform diverse textual inputs into fixed-length vectors that retain their inherent meaning and contextual information.

To integrate the semantic representation into the recommendation process, we employ a linear layer with dropout. This step serves to reduce the dimensionality of the large language model-enhanced semantic features and facilitates the mapping of semantic information into the collaboration space. The formal expression for this process is as follows:

$$\tilde{s_i} = \text{Linear}(s_i), \tag{6}$$

where $s_i \in \mathrm{R}^{1 \times d_{\text{LLM}}}$ is the input feature and $\tilde{s_i} \in \mathrm{R}^{1 \times d}$ is the output feature after projection.

4.3 Recommendation Module

Item Representation Fusion. We introduce a gate layer to integrate the semantic representation $\tilde{s_i}$ and collaborative representation n_i of the item i:

$$\mathbf{e}_i = \beta_i \cdot \tilde{s_i} + (1 - \beta_i) \cdot n_i, \tag{7}$$

$$\beta_i = \sigma(\mathbf{W}_{g_i}[\tilde{s_i}; n_i]), \tag{8}$$

where $[\cdot;\cdot]$ denotes concatenation, and $\mathbf{W}_{g_i}$ is learnable parameters.

User Representation Fusion. We extract the following two parts from the user-system conversation history as user preferences $\mathbf{e}_u$: utterances in conversation $[d_1, d_2, \cdots, d_n]$ and mentioned entities $[e_1, e_2, \cdots, e_n]$. Similar to calculating conversation history $\mathbf{h}$ in Sect. 4.2, we set up a self-attention layer to aggregate entity representations mentioned in the conversation:

$$\begin{aligned} \mathbf{u} &= \mathbf{E} \cdot \beta, \\ \beta &= \text{softmax}(\boldsymbol{b}_e^T \tanh(\mathbf{W}_e \mathbf{E})), \end{aligned} \tag{9}$$

where $\mathbf{E}$ denotes the representation matrix obtained by stacking the sequence of entity representations, $\mathbf{W}_c$ and $\boldsymbol{b}_c$ denote the learnable parameter matrix and vector of the self-attention layer.

To integrate the acquired user conversation history $\mathbf{h}$ and user interaction history $\mathbf{u}$, we establish a gate layer to derive the final user representation $\mathbf{e}_u$:

$$\mathbf{e}_u = \beta_u \cdot \mathbf{h} + (1 - \beta_u) \cdot \mathbf{u}, \tag{10}$$

$$\beta_u = \sigma(\mathbf{W}_{g_u}[\mathbf{h}; \mathbf{u}]), \tag{11}$$

where $\mathbf{W}_{g_u}$ is learnable parameters.

After learning the user's preference vector, we can calculate the user's potential rating for each item in the item set:

$$P_{rec}(u, i) = \text{softmax}(\mathbf{e}_u^T \cdot \mathbf{e}_i). \tag{12}$$

We use the cross-entropy loss to calculate the loss of the item rating given by the recommendation module and the item mentioned in the actual response. The formal expression is as follows:

$$\mathcal{L}_{rec} = -\frac{1}{-}N \sum_{i=1}^{N} \log(\hat{p}_i), \tag{13}$$

where N is the total number of recommendations, and $\hat{p}_i$ represents the probability that the target item is recommended.

To ensure that the learned semantic representation is beneficial for recommendation and to reduce the semantic gap between the LLMs-enhanced semantic representation and the CF-side rational representation, we incorporated similarity loss and contrastive loss as regularization terms in the training process:

$$\mathcal{L}_{sim} = -\log(\text{softmax}(e_u^T \cdot \tilde{s}_i)), \tag{14}$$

$$\mathcal{L}_{cont} = -\mathbb{E}\big(\log(\frac{\text{sim}(\tilde{s}_i, n_i)}{\sum_{j \in \mathcal{I}} \text{sim}(\tilde{s}_j, n_i)})\big). \tag{15}$$

The final objective function for the recommendation subtask is:

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda \mathcal{L}_{sim} + \gamma \mathcal{L}_{cont}. \tag{16}$$

4.4 Conversation Module

In this section, we focus on how to create response utterances that are both easy to understand and flow naturally. We achieve this objective by employing a modified version of the Transformer model. While the encoder component adheres to the conventional Transformer architecture, we introduce modifications to the decoder component. More precisely, we stack the representations of mentioned entities $[e_1, e_2, \cdots, e_n]$ and the semantic representations of referenced items $[i_1, i_2, \cdots, i_n]$ in the conversation, forming matrices $\mathbf{E}$ and $\mathbf{S}$. After the self-attention layer, we introduce two attention layers that integrate both the structural information of KG and the semantic information of items into the context representation:

$$\mathbf{A}_0^n = \text{MHA}(\mathbf{R}^{n-1}, \mathbf{R}^{n-1}, \mathbf{R}^{n-1}), \tag{17}$$

$$\mathbf{A}_1^n = \text{MHA}(\mathbf{A}_0^n, \mathbf{E}, \mathbf{E}), \tag{18}$$

$$\mathbf{A}_2^n = \text{MHA}(\mathbf{A}_1^n, \mathbf{S}, \mathbf{S}), \tag{19}$$

$$\mathbf{A}_3^n = \text{MHA}(\mathbf{A}_2^n, \mathbf{C}, \mathbf{C}), \tag{20}$$

$$\mathbf{R}^n = \text{FFN}(\mathbf{A}_3^n), \tag{21}$$

where $\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ represents the multi-head attention function [18], which takes query, key, and value as input:

$$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [h_1; \cdots; h_n]\mathbf{W}_A, \tag{22}$$

$$h_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^{\mathbf{Q}}, \mathbf{K}\mathbf{W}_i^{\mathbf{K}}, \mathbf{V}\mathbf{W}_i^{\mathbf{V}}), \tag{23}$$

where $[\cdot;\cdot]$ denotes the concatenation operation, n is the number of heads, $\mathbf{W}_A$ is a learnable matrix. $\text{FFN}(x)$ is a two level fully-connected feed-forward network with ReLU activation function:

$$\text{FFN}(x) = \text{ReLU}(x\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2. \tag{24}$$

In the above equations, $\mathbf{R}^n$ is the embedding matrix from the decoder at n-th layer, $\mathbf{C}$ is the embedding matrix output by the conversation encoder and $\mathbf{A}_0^n, \mathbf{A}_1^n, \mathbf{A}_2^n, \mathbf{A}_3^n$ are the output after self-attention, cross-attention with item-oriented KG, cross-attention with items' semantic representations and cross-attention with conversation context, respectively. Additionally, we incorporated the copy mechanism [23] to generate more meaningful utterances.

The loss function for the conversation subtask is defined as follows:

$$\mathcal{L}_{gen} = -\frac{1}{|C|}\sum_{t=1}^{|C|} log(P(d_t|d_1, \cdots, d_{t-1})) \tag{25}$$

where $|C|$ represents the number of turns in a conversation C and we compute this loss for each utterance d_t within C.

5 Experiment

To verify the superiority of our proposed model, we conduct extensive experiments to answer the following research questions: **RQ1**: How does our proposed model perform on recommendation and conversation tasks compared with existing methods? **RQ2**: How much does each component of our proposed model contribute to its overall performance? **RQ3**: Does our proposed item profile generation paradigm reduce the impact of noisy text on recommendation performance compared to directly utilizing item-related textual information? **RQ4**: How does using different LLM to encode item-related textual information affect the recommendation performance? **RQ5**: How do different hyperparameter settings affect the performance of our model?

5.1 Experiment Setup

Dataset. We conduct our experiments on Redial [11] and TG-Redial [24] datasets. The Redial dataset is a widely used English movie conversational recommendation dataset that comprises 10,006 dialogues, involving 504 users and 51,699 movies. The TG-ReDial dataset is a Chinese conversational recommendation dataset that comprises 10,000 dialogues, involving 1482 users and 33,834 movies. To conduct our experiments, we randomly divide the dataset into a training set, a validation set and a test set, in 8:1:1 ratio.

Baselines. Since we mainly evaluate the performance of CRS models on recommendation and conversation tasks, we consider the following existing work as our baseline:

- Popularity [7]: This model generates a recommendation list for a user based solely on how frequently items appear in the corpus.
- BERT [3]: This model utilizes BERT, a pretraining [20] model, to encode the user's current utterances and provide recommendation results.
- TextCNN [8]: This model utilizes a convolutional neural network approach to process the input conversation data and extract user representation vectors.
- Transformer [18]: This model utilizes an encoder-decoder approach based on the Transformer architecture to generate conversation responses.
- ReDial [11]: This model comprises an autoencoder-based recommendation module alongside a conversation module based on the HRED architecture.
- KBRD [1]: This model incorporates the DBpedia into CRS to improve the representation of recommended items.
- KGSF [23]: This model introduces the DBpedia to enhance the item representation and utilizes the ConceptNet to enhance the word representation.
- RevCore [12]: This model improves the performance of the CRS by injecting user reviews with similar sentiments into the dialogue.
- VRICR [21]: This model achieves a more accurate user representation by introducing resampling conversation-specific subgraphs based on the variational Bayesian method.

Table 1. The recommendation performance.

Dataset	Redial									TG-Redial								
Metric	HR			MRR			NDCG			HR			MRR			NDCG		
Method	@1	@5	@10	@1	@5	@10	@1	@5	@10	@1	@5	@10	@1	@5	@10	@1	@5	@10
Popularity	0.0111	0.0359	0.0536	0.0111	0.0196	0.0220	0.0111	0.0236	0.0294	0.0004	0.0004	0.0031	0.0004	0.0004	0.0008	0.0004	0.0004	0.0013
TextCNN	0.0114	0.0359	0.0710	0.0114	0.0199	0.0243	0.0114	0.0239	0.0349	0.0024	0.0087	0.0131	0.0024	0.0046	0.0051	0.0024	0.0056	0.0070
Bert	0.0248	0.0900	0.1437	0.0248	0.0470	0.0540	0.0248	0.0576	0.0748	0.0002	0.0018	0.0034	0.0002	0.0007	0.0009	0.0002	0.0010	0.0015
Redial	0.0297	0.1035	0.1659	0.0297	0.0533	0.0589	0.0297	0.0637	0.0842	0.0003	0.0020	0.0037	0.0003	0.0008	0.0010	0.0003	0.0011	0.0016
KBRD	0.0338	0.1147	0.1764	0.0338	0.0610	0.0691	0.0338	0.0742	0.0941	0.0055	0.0169	0.0243	0.0055	0.0095	0.0104	0.0055	0.0113	0.0137
KGSF	0.0335	0.1164	0.1795	0.0335	0.0614	0.0698	0.0335	0.0750	0.0953	0.0054	0.0193	0.0295	0.0054	0.0100	0.0114	0.0054	0.0123	0.0156
RevCore	0.0398	0.1229	0.1886	0.0398	0.0692	0.0776	0.0398	0.0834	0.1039	0.0061	0.0191	0.0302	0.0061	0.0104	0.0119	0.0061	0.0126	0.0162
VRICR	0.0434	0.1249	0.1901	0.0434	0.0703	0.0797	0.0434	0.0847	0.1055	0.0053	0.0183	0.0290	0.0053	0.0098	0.0113	0.0053	0.0119	0.0154
LGCRS\S	0.0384	0.1184	0.1809	0.0384	0.0658	0.0741	0.0384	0.0788	0.0989	0.0055	0.0198	0.0322	0.0055	0.0101	0.0117	0.0055	0.0125	0.0164
LGCRS\C	0.0417	0.1217	0.1860	0.0417	0.0688	0.0773	0.0417	0.0818	0.1026	0.0060	0.0219	0.0342	0.0060	0.0110	0.0126	0.0060	0.0137	0.0177
LGCRS	**0.0445**	**0.1254**	**0.1909**	**0.0445**	**0.0720**	**0.0807**	**0.0445**	**0.0852**	**0.1063**	**0.0067**	**0.0227**	**0.0357**	**0.0067**	**0.0122**	**0.0138**	**0.0067**	**0.0148**	**0.0189**

Evaluation Metrics. We measure the performance of our recommendation and conversation modules separately. When evaluating the recommendation module, our primary focus lies in assessing the CRS's ability to accurately reflect user preferences within the Top-K recommendation list. To assess the performance of the recommendation module, we utilize metrics such as HR@k, MRR@k, and NDCG@k. Considering the limited length of response in dialogues, we use a small k value ($k \in [1, 5, 10]$) to evaluate the performance of the recommendation module, which is more suitable for practical application scenarios. When evaluating the conversation module, we assess both the accuracy and diversity of the generated response. Accuracy is measured by indicators such as BLEU-1,2,3,4. To evaluate diversity, we use Distinct n-gram ($n = 3, 4$) at the sentence level.

Implementation Details. We build upon the CRSLab [22] and implement our methodology. The hidden layer representations for the recommendation and conversation modules are both set to a dimension of 300. When aggregating information from the knowledge graph using GNN-based method, we set the depth to 1. The hyperparameters λ and γ in the recommendation module are set to 1.0 and 0.06, respectively. They are determined through grid search on the validation set. We choose vicuna-13b-v1.5-16k [2] as the LLM backend and utilize the Adam optimizer with default parameter settings.

5.2 Overall Performance (RQ1)

Recommendation. We perform several experiments to validate the effectiveness of our proposed model for the recommendation task. We present the results of various recommendation methods on distinct recommendation metrics in Table 1. According to the results, we can find that the conversational recommendation method (e.g., Redial, KBRD, KGSF) is significantly better than the traditional recommendation method (e.g., Popularity, TextCNN, Bert). This observation underscores the essential role of contextual information in the conversational recommendation scenario. Moreover, approaches incorporating external knowledge exhibit superior performance compared to those that do

Table 2. The conversation performance.

Dataset	Redial						TG-Redial					
Metric	BLEU@1	BLEU@2	BLEU@3	BLEU@4	DIST@3	DIST@4	BLEU@1	BLEU@2	BLEU@3	BLEU@4	DIST@3	DIST@4
Transformer	0.0983	0.0222	0.0114	0.0063	0.2258	0.3867	0.2667	0.0440	0.0145	0.0065	1.7392	2.7114
Redial	0.0409	0.0043	0.0007	0.0002	0.3064	0.7086	0.1177	0.0158	0.0021	0.0004	0.7564	1.1854
KBRD	0.1712	0.0314	0.0135	0.0085	0.3476	0.6231	0.2617	0.0446	0.0151	0.0075	2.3076	3.3234
KGSF	0.1666	0.0334	0.0176	0.0109	0.4458	1.0366	0.2632	0.0473	0.0134	0.0061	2.2022	3.4458
RevCore	0.1121	0.0327	0.0196	0.0113	0.4822	0.6064	0.2432	0.0490	0.0170	0.0090	1.7048	2.9774
VRICR	0.1036	0.0241	0.0099	0.0045	0.5129	0.7323	0.2274	0.0491	0.0177	0.0085	2.6649	4.6301
LGCRS	**0.1757**	**0.0418**	**0.0227**	**0.0139**	**0.5622**	**1.3600**	**0.2721**	**0.0510**	**0.0181**	**0.0092**	**3.7480**	**6.0010**

not use external knowledge, highlighting the effectiveness of incorporating item knowledge to enhance recommendation performance. Most notably, our model demonstrates superior performance when compared to all baseline methods in the recommendation task. This substantiates the effectiveness of our method in enhancing item representation through the utilization of an item profile generated by a large language model.

Conversation. We validate the performance of our proposed model on conversation task in terms of both accuracy and diversity, compare with baselines and conduct in-depth analysis. We show the performance of different conversation methods on conversation tasks in Table 2. Our model outperforms all the other models on both metrics, with the highest BLEU and DIST scores across all n-gram levels. The reason is that our model utilizes large language models to enhance item representation to reduce irrelevant information, thereby producing responses closer to ground truth. Overall, our model is capable of generating diverse responses while preserving similarity to ground truth responses.

5.3 Ablation Study (RQ2)

In this section, we conduct an ablation study to assess the contribution of each component to recommendation performance. The results are presented in Table 1. We consider two variants of our method: (1) LGCRS\S removes the semantic information contained in the item representation. (2) LGCRS\C removes the alignment constraints of the semantic representation and the collaborative filtering representation.

It is apparent that each component plays a substantial role in recommendation task, as evidenced by the notable decline in recommendation performance observed in both variants. Particularly, when compared with LGCRS\C, LGCRS\S exhibits a greater performance decrease, suggesting that the item profile generated by integrating LLMs effectively enhances the understanding of items in recommender systems. Moreover, LGCRS\C leads to performance degradation, indicating that LGCRS effectively reduce the semantic gap between the LLMs-enhanced semantic representation and the CF-side rational representation.

Table 3. Text Enhance Method Analysis.

Dataset	Redial									TG-Redial								
Method	H@1	H@5	H@10	M@1	M@5	M@10	N@1	N@5	N@10	H@1	H@5	H@10	M@1	M@5	M@10	N@1	N@5	N@10
sep.	0.0415	0.1174	0.1856	0.0415	0.0673	0.0763	0.0415	0.0796	0.1016	0.0058	0.0192	0.0299	0.0058	0.0101	0.0114	0.0058	0.0123	0.0156
agg.	0.0417	0.1197	0.1809	0.0417	0.0689	0.0770	0.0417	0.0814	0.1012	0.0058	0.0223	0.0352	0.0058	0.0112	0.0129	0.0058	0.0139	0.0181
enhance	0.0445	0.1254	0.1909	0.0445	0.0720	0.0807	0.0445	0.0852	0.1063	0.0067	0.0227	0.0357	0.0067	0.0122	0.0138	0.0067	0.0148	0.0189

5.4 Text Enhance Method Analysis (RQ3)

In this section, we investigate the influence of employing various methods to introduce item-related text information on recommendation performance. We primarily focus on the following three methods:

- sep.: This method employs LLMs to generate a representation for each review associated with the item. Subsequently, the generated representations are fed into the self-attention layer to derive the semantic representation of the item.
- agg.: This method directly utilizes LLMs to encode concatenated item reviews, thereby obtaining item semantic representations.
- enhance: This method initially employs LLMs to refine item profiles based on descriptions generated from reviews, utilizing manually designed prompts. Subsequently, a text encoding model is employed to obtain semantic representations of the items.

In Table 3, we showcase the performance on recommendation task using vicuna-13b-v1.5-16k as the backend for text encoding on two datasets. The results reveal that the performance of the 'enhance' method surpasses that of 'agg.' and 'sep.'. This can be attributed to the following two factors: 1) Integration of Information: Compared with 'sep.' that considers each item review separately, 'agg.' leverages the excellent natural language understanding capability of the large language model. It integrates the information contained in reviews more effectively than the simple self-attention layer of 'sep.'. 2) Logical Reasoning: With manually designed prompts, the 'enhance' method utilizes the logical reasoning ability of the large language model to reduce noise in the review data, thereby enhancing the quality of the item profile. In summary, our proposed item profile generation paradigm effectively reduces noise and improves recommendation performance compared to directly utilizing item-related textual information.

5.5 LLM Backend Analysis (RQ4)

In this section, we investigate the impact of employing different large language model for processing text information on recommendation performance. We consider the utilization of the following five open-source large language models: chatglm3-6b [4], baichuan2-7b-chat [19], baichuan2-13b-chat, vicuna-7b-v1.5-16k [2] and vicuna-13b-v1.5-16k. In the Table 4, we present the results of employing the 'enhance' method to process item-related text information for the recommendation task on the Redial dataset.

Table 4. LLM Backend Analysis.

LLM Backend	H@1	H@5	H@10	M@1	M@5	M@10	N@1	N@5	N@10
chatglm3-6b	0.0410	0.1206	0.1844	0.0410	0.0683	0.0767	0.0410	0.0812	0.1017
baichuan2-7b	0.0420	0.1184	0.1886	0.0420	0.0685	0.0777	0.0420	0.0809	0.1034
baichuan2-13b	0.0426	0.1213	0.1832	0.0426	0.0695	0.0777	0.0426	0.0823	0.1022
vicuna-7b	0.0422	0.1234	0.1846	0.0422	0.0699	0.0781	0.0422	0.0830	0.1029
vicuna-13b	0.0445	0.1254	0.1909	0.0445	0.0720	0.0807	0.0445	0.0852	0.1063

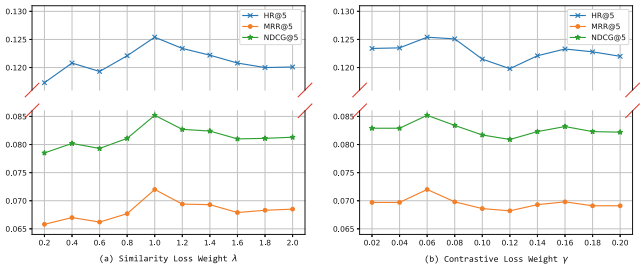

Fig. 2. Hyper-parameters sensitive analysis on the Redial dataset.

From the results, it is evident that whether using baichuan2 or vicuna-v1.5, the recommendation outcomes corresponding to the 13b model are marginally superior to those of the 7b model. The recommendation performance using vicuna-v1.5 as the backend surpasses that of chatglm3 and baichuan2. This difference in performance may be attributed to the fact that the latter two models have undergone fine-tuning on Chinese conversation data, resulting in a partial loss of English understanding and processing capabilities.

5.6 Parameter Sensitive Analysis (RQ5)

In this section, we examine the sensitivity of two hyperparameters: the contrast loss weight λ and the similarity loss weight γ. For each hyperparameter, we explore its values within the empirical space and report ten representative values in Fig. 2. In general, as the weight increases, the recommendation performance initially rises and subsequently descends. In a word, employing regularization terms to align the semantic representation space and collaborative signal space of items proves beneficial in enhancing the recommendation performance of conversational recommender systems, and we should tune them in fine-grained.

6 Conclusion

In this paper, we introduce a novel method LGCRS that utilizes item semantic representations to improve recommendation performance and generate more

natural responses. Through the use of a large language model to refine user reviews, we have obtained a high-quality item profile for generating semantic representations. LGCRS enhances item understanding and improves recommendation effectiveness by constructing a multi-aspect item representation that integrates semantic representations with CF-side rational representations. Experiments on publicly available conversational recommendation datasets show that our method performs better in both recommendation and conversation tasks.

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